Exceptional sets for sums and differences of two primes along polynomial sequences

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Abstract

For a fixed integer polynomial Q of degree $d = 2$ or 3 with positive leading coefficient, we study the parameters $1 \le m \le N$ for which $Q(m)$ is positive and even but is not a sum of two primes. Assuming the fourth-moment and major-arc theorems of Li–Liu under a common fixed Dirichlet zero-free boundary $\Theta < 11/12$, we obtain $O_{Q,\varepsilon}(N^{d/4+\varepsilon})$ exceptions. The same bound holds for failure of a prime-difference representation with both primes at most $C_Q N^d$, where C_Q is explicit. Pintz's major-arc formula and the exponential-sum estimate of Maynard–Pandey–Radziwiłł, under a common boundary $\Theta < 1$, give the alternative bound $O_{Q,\varepsilon}(N^{7d/24+\varepsilon})$. For quadratic polynomials mapping an entire integer interval into $[X/2, X]$, the corresponding bounds are $O_\varepsilon(X^{1/4+\varepsilon})$ and $O_\varepsilon(X^{7/24+\varepsilon})$, uniformly in the coefficients and the interval. OpenAI's seven-eighths theorem supplies the zero-free hypothesis in both routes. The argument is the classical thin-set exceptional-set mechanism, with the exponential sum taken over the exceptions themselves; for $Q(t) = 2t^2$ the alternative route is the argument of Perelli–Pintz with this restriction. We give the elementary multiplicity arguments, including the cubic zero branches.

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1 Statements and context

All logarithms are natural, $e(t) = \exp(2\pi i t)$, and primes may repeat. Parameters $m \le N$ always mean positive integers $1 \le m \le N$. Let $Q \in \mathbb{Z}[t]$ have degree $d \in \{2, 3\}$ and positive leading coefficient. For $N, Z \ge 2$ define

$$E_+(N; Q) = \#\{1 \le m \le N : Q(m) > 0,\ 2 \mid Q(m),\ Q(m) \ne p_1 + p_2\}, \tag{1.1}$$

$$E_-(N; Q; Z) = \#\{1 \le m \le N : Q(m) > 0,\ 2 \mid Q(m), \\ Q(m) \ne p_2 - p_1 \text{ for primes } p_1, p_2 \le Z\}. \tag{1.2}$$

The coefficients of a fixed polynomial are independent of N. Set

$$A_Q = \sum_{j=0}^{d} \left|[t^j]Q\right|, \qquad C_Q = 2(1 + A_Q). \tag{1.3}$$

We use the notation $\mathrm{ZF}(\Theta)$ for the following common zero-free hypothesis: every nontrivial zero of every primitive Dirichlet L-function, including the character of conductor one, has real part at most the same fixed number Θ. The boundary is uniform in conductor and height. A principal pole at $s = 1$ is allowed. We state all analytic inputs in Section 2.

Theorem 1.1 (Conditional on the Li–Liu inputs)**.** *Assume* $\mathrm{ZF}(\Theta)$ *for a fixed* $1/2 \le \Theta < 11/12$*, and use Li–Liu's Theorem 1 and Proposition 4.1 as stated in Inputs 2.3 and 2.4. For every fixed* Q *as above and every* $\varepsilon > 0$*,*

$$E_+(N;Q) + E_-(N;Q;C_Q N^d) \ll_{Q,\Theta,\varepsilon} N^{d/4+\varepsilon}. \tag{1.4}$$

Thus the parameter exponents are $1/2$ *for quadratics and* $3/4$ *for cubics. No irreducibility hypothesis or additional local condition on* Q *is required.*

Theorem 1.2 (Conditional Pintz–MPR alternative)**.** *Assume* $\mathrm{ZF}(\Theta)$ *for a fixed* $1/2 \le \Theta < 1$*, and use Pintz's Theorem 1, (2.10)–(2.11), and Maynard–Pandey–Radziwiłł's Theorem 1.1 as stated in Inputs 2.5 and 2.6. For every fixed* Q *as above and every* $\varepsilon > 0$*,*

$$E_+(N;Q) + E_-(N;Q;C_Q N^d) \ll_{Q,\Theta,\varepsilon} N^{7d/24+\varepsilon}. \tag{1.5}$$

The parameter exponents are $7/12$ *and* $7/8$*, respectively.*

Here and below constants may also depend on the fixed weight and the fixed constants of the cited analytic inputs. The alternative theorem uses a different prime sum and a different choice of arcs from Theorem 1.1.

For a finite interval I of consecutive integers and a quadratic $Q \in \mathbb{Z}[t]$, put

$$\begin{aligned} U_+(I;Q) &= \#\{m \in I : 2 \mid Q(m),\ Q(m) \neq p_1 + p_2\}, \\ U_-(I;Q;Z) &= \#\{m \in I : 2 \mid Q(m),\ Q(m) \neq p_2 - p_1 \text{ with } p_1, p_2 \le Z\}. \end{aligned}$$

The range condition in the next theorem makes all these values positive. Perelli [10, p. 478] leaves estimates uniform in the coefficients as a separate question.

Theorem 1.3 (Conditional uniform quadratic bound)**.** *Let* $X \ge 2$*, let* I *be a finite interval of consecutive integers, and let* $Q \in \mathbb{Z}[t]$ *have degree exactly two. Suppose that*

$$Q(m) \in [X/2, X] \qquad (m \in I). \tag{1.6}$$

Under the inputs of Theorem 1.1,

$$U_+(I;Q) + U_-(I;Q;2X) \ll_{\Theta,\varepsilon} X^{1/4+\varepsilon}. \tag{1.7}$$

Under the inputs of Theorem 1.2,

$$U_+(I;Q) + U_-(I;Q;2X) \ll_{\Theta,\varepsilon} X^{7/24+\varepsilon}. \tag{1.8}$$

The constants are independent of the coefficients of Q*, the location of* I*, and its length. Either sign separately is also bounded by* $|I|$*. The leading coefficient may have either sign.*

Corollary 1.4 (Specialization of the zero-free input)**.** *OpenAI's Theorem 1.1, stated in Input 2.2, gives* $\mathrm{ZF}(7/8)$*. Hence the conclusions of Theorems 1.1–1.3 hold with* $\Theta = 7/8$*, relative to the remaining inputs of each theorem (Inputs 2.3–2.4, respectively Inputs 2.5–2.6). In particular this applies to* $Q(t) = 2t^2$ *and* $Q(t) = 2t^3$ *and to their bounded prime-difference versions.*

1.1 The classical thin-set mechanism

If $\mathcal{A}$ is the set of distinct exceptional polynomial values and $E = |\mathcal{A}|$, the exponential sum $K(\alpha) = \sum_{a\in\mathcal{A}} e(a\alpha)$ retains the exceptional set itself. A bound $r_{\mathcal{A}}(s) \le D$ for ordered pair-sum multiplicities gives

$$\int_{\mathbb{T}} |K(\alpha)|^4 \, d\alpha \le DE^2.$$

The major coefficients are of size at least X, whereas the full coefficients vanish at the exceptional values. Hölder's inequality therefore transfers either a fourth moment or a pointwise bound for the prime sum to a bound for E. Brüdern–Kawada–Wooley [2, Introduction] describe working with an exponential sum over the exceptions. In the proof of their Theorem 1 [2, §2, (5)], as in Perelli–Pintz [11, (3.6)–(3.10)], this is the sign-weighted sum over all parameters, whose moments are bounded by the full polynomial mean value. Here the sum runs over the exceptional values only, so that its fourth moment is at most DE^2. The elementary part of the present argument supplies $D \ll X^{\eta}$ for fixed quadratics and cubics, and uniformly for the quadratic families in (1.6). Applying the two analytic interfaces gives the bounds in Table 1.

Analytic route	Physical block	Fixed degree 2	Fixed degree 3
Li–Liu	$X^{1/4+\varepsilon}$	$N^{1/2+\varepsilon}$	$N^{3/4+\varepsilon}$
Pintz–MPR	$X^{7/24+\varepsilon}$	$N^{7/12+\varepsilon}$	$N^{7/8+\varepsilon}$

Table 1: Bounds for the number of exceptional parameters, relative to the inputs in the corresponding theorem. A physical block means $Q(m) \in [X/2, X]$; for fixed Q its largest scale is $O_Q(N^d)$. The physical bounds are uniform in the quadratic setting of Theorem 1.3.

The difference problem uses the Fourier coefficients of $|S|^2$ in place of S^2. The fourth moment of K still counts pair *sums* of exceptional values. This observation allows the same polynomial multiplicity lemmas to treat both signs.

1.2 Related work

Perelli [10, Theorem 1 and Corollary 1] treats a fixed polynomial $F \in \mathbb{Z}[t]$ of degree k with positive leading coefficient. Writing his physical scale as Y, for $Y^{1/(3k)+\varepsilon} \le H \le Y^{1/k}$ his Corollary 1 gives, among the parameters $Y^{1/k} \le m \le Y^{1/k} + H$ with even $F(m)$, $O_{A,\varepsilon,F}(H(\log Y)^{-A})$ Goldbach exceptions, for every $A > 0$. His Theorem 1 is the underlying mean-square estimate for the weighted representation function. Brüdern–Kawada–Wooley [2, Theorem 1] prove that, for a fixed positive-leading degree-k polynomial Φ, the number of $m \le N$ for which $2\Phi(m)$ is not a sum of two primes is $O_{\Phi}(N^{1-c/k})$, where $c > 0$ is absolute. Their introduction describes the use of an exponential sum over the exceptions; in the proof of Theorem 1 this is a sign-weighted sum over all parameters [2, §2, (5)]. A polynomial whose even values occur only on some residue classes is accommodated by splitting m modulo two: whenever $Q(r)$ is even, $Q(2t + r)/2 \in \mathbb{Z}[t]$.

For doubled squares write

$$E_2(N) = \#\{1 \le m \le N : 2m^2 \ne p_1 + p_2\}.$$

Pintz [12, Theorem 5], in a theorem attributed there to Perelli and Pintz, states, unconditionally, $E_2(N) \ll_{\varepsilon} N^{4/5+\varepsilon}$. Perelli–Pintz [11, Theorem 1 and (1.4)] state $E_2(N) \ll_{\varepsilon} N^{19/24+\varepsilon}$, using the fixed Dirichlet zero-free half-plane together with Pintz's major arcs and MPR's exponential-sum estimate. Here N is the parameter cutoff for $2m^2$. In physical size $2m^2 \le X$, their stated bound

becomes $X^{19/48+\varepsilon}$. For $Q(t) = 2t^2$ the proof of Theorem 1.2 is the argument of [11, (3.1)–(3.10)], with the same inputs and the same Hölder inequality, except that the sign-weighted sum over all $m \le N$ is replaced by the sum over the exceptional values. Its fourth moment is then $O(E^2N^\varepsilon)$ instead of $O(N^{2+\varepsilon})$, which gives $N^{7/12+\varepsilon}$ in place of $N^{19/24+\varepsilon}$. Theorem 1.1 gives $N^{1/2+\varepsilon}$ by using the Li–Liu fourth moment in place of the pointwise bound.

Pintz–Althöfer [14, Theorem 1 and Corollary 2] prove $E(X) \ll_\varepsilon X^{7/12+\varepsilon}$ for the ordinary Goldbach exceptional set under any common fixed Dirichlet zero-free boundary below one, and insert OpenAI's seven-eighths theorem. Their Sections 2.1–2.2 align Pintz's major arcs with the MPR bound. Li–Liu [4, Theorems 1–2 and Proposition 4.1] obtain $E_+(X), E_-(X) \ll_\Theta X^{1/2}(\log X)^{100}$ under ZF(Θ) for $\Theta < 11/12$; their new analytic estimate is the minor-arc fourth moment. We use that estimate together with their major coefficients, preserving its weight and arc parameter.

Later work lowers the zero-free boundary below 7/8 (for instance [1, 3, 5, 6, 16]). The QRH tracker records the Dirichlet boundary 0.8739558 [15]. Within the two routes of this paper such improvements leave every exponent in Table 1 unchanged: in the Li–Liu parameter selection Θ occurs through $\max(\Theta, 7/8)$, and in the alternative route the term $X^{19/24}$ in the MPR estimate determines the power 7/24.

Section 2 records the inputs and their compatibility. Section 3 proves the finite transfer, and Section 4 supplies the algebraic multiplicity bounds. Sections 5–6 prove the main results. The appendix describes the scope of the ancillary checks.

2 Analytic inputs

Hypothesis 2.1 (Common Dirichlet zero-free boundary)**.** For one fixed $1/2 \le \Theta < 1$, every nontrivial zero ρ of every primitive Dirichlet L-function satisfies $\operatorname{Re}\rho \le \Theta$. This includes ζ, the L-function of conductor one, and applies at all conductors and all heights. Denote this assertion by ZF(Θ).

Input 2.2 (OpenAI, Theorem 1.1, Dirichlet specialization)**.** *Every Dirichlet L-function, including ζ, has no zero in $\operatorname{Re} s > 7/8$. A pole at $s = 1$ for a principal character is allowed. This is the Dirichlet assertion of [8, Theorem 1.1].*

References to the OpenAI manuscript and its Lean scope document [8, 9] are pinned to the repository commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The full Theorem 1.1 also treats finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$. The Dirichlet assertion alone supplies Hypothesis 2.1 at $\Theta = 7/8$. The Dirichlet comparator linked by the family 003 scope document uses the declaration name

`OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re`.

We use the written theorem with this scope. The ancillary computations below verify finite algebra and examples.

2.1 The Li–Liu weight, arcs, and theorems

Fix a real function $w \in C_c^\infty((0,1))$ such that $0 \le w \le 1$ and $w = 1$ on $[1/8, 7/8]$. Choose it in the logarithmic Gevrey class of order two specified in [4, §2]: for fixed constants $C_0, C_1 > 0$ and all integers $j \ge 0$,

$$\sup_{v\in\mathbb{R}} \left| \frac{d^j}{dv^j} w(e^v) \right| \le C_0 C_1^j (j!)^2. \tag{2.1}$$

The weight is fixed once and for all. Put

$$S_X(\alpha) = \sum_p (\log p) w(p/X) e(p\alpha). \tag{2.2}$$

On $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ use Lebesgue measure of total mass one. Write $\|t\|_{\mathbb{T}}$ for the distance from t to the nearest integer, and define

$$\mathfrak{M}_X(P) = \bigcup_{q \le P} \bigcup_{\substack{a \bmod q \\ (a,q)=1}} \left\{ \alpha : \|\alpha - a/q\|_{\mathbb{T}} \le \frac{P}{qX} \right\}, \qquad \mathfrak{m}_X(P) = \mathbb{T} \setminus \mathfrak{M}_X(P). \tag{2.3}$$

For a fixed exponent $\kappa < 1/2$ and $P = X^\kappa$, the arcs are disjoint for sufficiently large X.

Input 2.3 (Li–Liu, Theorem 1)**.** *Assume* ZF(Θ) *for a fixed* $1/2 \le \Theta < 11/12$. *There is a fixed* $\kappa = \kappa(\Theta) \in (4/9, 9/20)$ *such that*

$$\int_{\mathfrak{m}_X(X^\kappa)} |S_X(\alpha)|^4 \, d\alpha \ll_{w,\Theta} X^{5/2} (\log X)^{100}. \tag{2.4}$$

The weight and arcs are those in (2.1)–(2.3).

For a nonzero even integer r define

$$\mathfrak{S}(r) = 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{\substack{p \mid r \\ p>2}} \frac{p-1}{p-2}. \tag{2.5}$$

The infinite product converges to a positive number, since $\sum_{p>2} (p-1)^{-2} < \infty$, and each factor in the finite product is at least one. Thus $\mathfrak{S}(r) \gg 1$ uniformly in nonzero even r. Define the finite weight convolutions

$$I_w^+(n) = \sum_{k \in \mathbb{Z}} w(k/X) w((n-k)/X), \quad I_w^-(h) = \sum_{k \in \mathbb{Z}} w(k/X) w((k+h)/X). \tag{2.6}$$

Input 2.4 (Li–Liu, Proposition 4.1)**.** *Assume* ZF(Θ) *for a fixed* $1/2 \le \Theta < 1$. *For every fixed* $0 < \kappa < 9/20$ *there is* $c_1 > 0$ *such that, with* $P = X^\kappa$,

$$\int_{\mathfrak{M}_X(P)} S_X(\alpha)^2 e(-n\alpha) \, d\alpha = \mathfrak{S}(n) I_w^+(n) + O_{w,\Theta,\kappa}(X^{1-c_1}), \tag{2.7}$$

$$\int_{\mathfrak{M}_X(P)} |S_X(\alpha)|^2 e(-h\alpha) \, d\alpha = \mathfrak{S}(h) I_w^-(h) + O_{w,\Theta,\kappa}(X^{1-c_1}), \tag{2.8}$$

uniformly for even $n \in [X/2, X]$ *and even* $2 \le h \le X/2$, *respectively. The weight is the same as in Input 2.3.*

The flat portion of w makes the main terms uniformly positive. For $n \in [X/2, X]$, the intersection $[X/8, 7X/8] \cap [n - 7X/8, n - X/8]$ has length at least $X/4$. For $2 \le h \le X/2$, the intersection $[X/8, 7X/8] \cap [X/8 - h, 7X/8 - h]$ has length $3X/4 - h \ge X/4$. Counting integers in these intervals gives $I_w^+(n), I_w^-(h) \ge X/4 - O(1)$. Consequently, for some fixed $c > 0$ and all sufficiently large X,

$$\operatorname{Re} \int_{\mathfrak{M}_X(P)} S_X^2 e(-n\alpha) \, d\alpha \ge cX, \tag{2.9}$$

$$\operatorname{Re} \int_{\mathfrak{M}_X(P)} |S_X|^2 e(-h\alpha) \, d\alpha \ge cX \tag{2.10}$$

on the stated ranges. In particular Input 2.4 applies to the same κ supplied by Input 2.3.

The proof of Li–Liu's Theorem 1 chooses κ as in [4, Proposition 9.4], which uses

$$\max\left\{\frac{1351}{3006}, \frac{5-24\delta}{10-40\delta}\right\} < \kappa < \frac{9}{20}, \qquad \delta = 1-\max(\Theta, 7/8) > \frac{1}{12}. \tag{2.11}$$

At $\Theta = 7/8$, taking $\kappa = 899/2000$ gives

$$\frac{(5/2)(1/2-\kappa)}{1+10(1/2-\kappa)} = \frac{101}{1204} < \frac{1}{8}, \qquad \frac{9}{2}\left(\frac{1}{2}-\kappa\right) = \frac{909}{4000} < \frac{38}{167}. \tag{2.12}$$

The last margin is $197/668000$. These rational inequalities are checked in `anc/exact_checks.py`. Theorem 1 of Li–Liu is used with its existential choice of κ; (2.11) records the narrower choice from its proof.

2.2 Pintz's major-arc formula and MPR's pointwise estimate

Fix a sufficiently small $\nu > 0$, with $\nu < 1/72$, and set

$$P_0 = X^{4/9-\nu}, \quad X_1 = X^{1-\rho}, \quad T_X(\alpha) = \sum_{X_1 < p \le X} (\log p) e(p\alpha). \tag{2.13}$$

Here $\rho > 0$ is a fixed sufficiently small admissible cutoff exponent chosen as in Pintz's theorem after ν. We keep this cutoff exponent separate from positive exponents in error terms.

Input 2.5 (Pintz, Theorem 1, (2.10)–(2.11), zero-free specialization)**.** *Use the numbering of [13, arXiv v1]. Assume* $\mathrm{ZF}(\Theta)$ *for a fixed* $1/2 \le \Theta < 1$. *With* (2.13)*, for each fixed sufficiently large* H *and all sufficiently large* X*, one can choose the same* $P \in [P_0 X^{-\nu}, P_0]$ *for the two formulas*

$$\int_{\mathfrak{M}_X(P)} T_X(\alpha)^2 e(-n\alpha)\, d\alpha = \mathfrak{S}(n) n + O_\nu(\mathfrak{S}(n) X e^{-c_\nu H}) + O_\nu(X^{1-c_2}), \tag{2.14}$$

$$\int_{\mathfrak{M}_X(P)} |T_X(\alpha)|^2 e(-h\alpha)\, d\alpha = \mathfrak{S}(h)(X - X_1 - h + O(1)) + O_\nu(\mathfrak{S}(h) X e^{-c_\nu H}) + O_\nu(X^{1-c_2}), \tag{2.15}$$

for even $n \in [X/2, X]$ *and even* $h \in [X/4, X/2]$*, respectively. Here* $c_\nu, c_2 > 0$ *are fixed, and the implicit constants may depend on the fixed parameters of the formula. The arcs are* (2.3) *at this same* P*.*

To specify this specialization, Pintz's set of generalized exceptional singularities consists of the principal pole and nontrivial zeros with conductor at most P, real part at least $1 - H/\log X$, and ordinate of absolute value at most $\sqrt{X}$. First fix H; then $1 - H/\log X > \Theta$ for large X, so only the pole remains. Its term in (2.10) is $\mathfrak{S}(n) n$, since $\Gamma(1)^2/\Gamma(2) = 1$. In (2.11) its integer convolution is

$$\#\{(k,\ell) \in (X_1, X]^2 \cap \mathbb{Z}^2 : k - \ell = h\} = X - X_1 - h + O(1).$$

The positive power c_2 may be decreased to absorb fixed logarithmic factors or the cutoff error. Choosing H large and then X large gives a common lower bound cX for the real parts of both major coefficients. This is a pointwise lower bound for every target in the displayed ranges. The sum specialization also appears in [14, §2.1].

Input 2.6 (Maynard–Pandey–Radziwiłł, Theorem 1.1)**.** *Let $Y \geq 2$, $q \in \mathbb{N}$, $(a,q) = 1$, $q \leq \sqrt{Y}$, and $|\alpha - a/q| \leq 1/(q\sqrt{Y})$. Put $B_Y = \max\{q, qY|\alpha - a/q|\}$. For every $\eta > 0$,*

$$\left|\sum_{n<Y} \Lambda(n)e(n\alpha)\right| \ll_\eta Y^\eta \left(\frac{Y}{\sqrt{B_Y}} + Y^{19/24}\right). \tag{2.16}$$

Here $\Lambda(p^j) = \log p$ for primes p and integers $j \geq 1$, and $\Lambda(n) = 0$ otherwise. This is the arbitrary-small-power form of [7, Theorem 1.1].

Changing $n < Y$ to $n \leq Y$ costs at most $\log(2Y)$. Theorem 1.2 uses the external estimate (2.16).

3 A finite exceptional-set transfer

For a finite set $\mathcal{A}$ of distinct integers write

$$r_{\mathcal{A}}(v) = \#\{(a,b) \in \mathcal{A}^2 : a + b = v\}. \tag{3.1}$$

These pairs are ordered. For $K(\alpha) = \sum_{a\in\mathcal{A}} e(a\alpha)$, finite orthogonality gives

$$\int_{\mathbb{T}} |K(\alpha)|^4 \, d\alpha = \sum_{v\in\mathbb{Z}} r_{\mathcal{A}}(v)^2 \leq \left(\sup_v r_{\mathcal{A}}(v)\right) |\mathcal{A}|^2. \tag{3.2}$$

Lemma 3.1 (Finite transfer)**.** *Let S be a trigonometric polynomial on $\mathbb{T}$, and let $\mathfrak{M}, \mathfrak{m}$ be complementary measurable subsets of $\mathbb{T}$. Let $\mathcal{A}$ be a finite set of distinct integers, $E = |\mathcal{A}|$, and $c, X > 0$. Choose either $F = S^2$ or $F = |S|^2$. Suppose that, for every $a \in \mathcal{A}$,*

$$\int_{\mathbb{T}} F(\alpha)e(-a\alpha)\, d\alpha = 0, \qquad \operatorname{Re}\int_{\mathfrak{M}} F(\alpha)e(-a\alpha)\, d\alpha \geq cX. \tag{3.3}$$

If $r_{\mathcal{A}}(v) \leq D$ for every integer v, put $B_j = \int_{\mathfrak{m}} |S|^j$ for $j = 2, 4$, and $M = \sup_{\mathfrak{m}} |S|$, with $M = 0$ when $\mathfrak{m}$ is empty. Then

$$E \leq c^{-2}X^{-2}D^{1/2} \min\{B_2 B_4^{1/2},\ M B_2^{3/2}\}. \tag{3.4}$$

Proof. If $E = 0$, the conclusion holds directly. Suppose $E > 0$, so $D \geq 1$, and put $K(\alpha) = \sum_{a\in\mathcal{A}} e(a\alpha)$. Summing the real parts of the major coefficients in (3.3) and using the vanishing full coefficients gives

$$cXE \leq \left|\int_{\mathfrak{m}} F(\alpha)\overline{K(\alpha)}\, d\alpha\right| \leq \int_{\mathfrak{m}} |S(\alpha)|^2 |K(\alpha)|\, d\alpha. \tag{3.5}$$

Both choices of F have modulus $|S|^2$. Hölder's inequality with exponents $2, 4, 4$, followed by (3.2), yields

$$cXE \leq B_2^{1/2} B_4^{1/4} \left(\int_{\mathbb{T}} |K|^4\right)^{1/4} \leq B_2^{1/2} B_4^{1/4} D^{1/4} E^{1/2}.$$

Alternatively, $|S|^2 \leq M^{1/2}|S|^{3/2}$ on $\mathfrak{m}$. Hölder with exponents $4/3, 4$ gives

$$cXE \leq M^{1/2} B_2^{3/4} \left(\int_{\mathbb{T}} |K|^4\right)^{1/4} \leq M^{1/2} B_2^{3/4} D^{1/4} E^{1/2}.$$

Squaring either inequality and cancelling c^2X^2E proves its part of (3.4). If $\mathfrak{m}$ is empty, the hypotheses force $E = 0$, which is covered by the first case. □

For either prime sum used here the second moment has the elementary bound

$$\int_{\mathbb{T}} \left| \sum_{p \le X} (\log p) a_p e(p\alpha) \right|^2 d\alpha = \sum_{p \le X} (\log p)^2 |a_p|^2 \le X(\log X)^2 \qquad (|a_p| \le 1). \tag{3.6}$$

Thus the only prime-distribution estimates used in the transfer are the major-arc lower bound and the chosen minor-arc input.

Corollary 3.2 (The two physical powers)**.** *Suppose the conditions of Lemma 3.1 hold, and for every $\eta > 0$ suppose $D \ll_\eta X^\eta$ and $B_2 \ll X(\log X)^2$, with the desired uniformity of constants. If $B_4 \ll_\eta X^{5/2+\eta}$, then $E \ll_\varepsilon X^{1/4+\varepsilon}$. If instead $M \ll_\eta X^{19/24+\eta}$, then $E \ll_\varepsilon X^{7/24+\varepsilon}$.*

Proof. The first expression on the right of (3.4) is

$$\ll_\eta X^{-2+\eta/2} X(\log X)^2 X^{5/4+\eta/2} = X^{1/4+\eta}(\log X)^2.$$

The second is

$$\ll_\eta X^{-2+\eta/2} X^{19/24+\eta} X^{3/2} (\log X)^3 = X^{7/24+3\eta/2}(\log X)^3.$$

Choose η sufficiently small relative to ε, then absorb the fixed logarithmic factors into the remaining small power. □

The set $\mathcal{A}$ in this application consists of the actual distinct exceptions. Its cardinality E enters the fourth moment as E^2, preserving the arithmetic information used in the classical method.

4 Polynomial pair-sum multiplicities

4.1 Divisors and quadratic polynomials

Lemma 4.1 (Elementary divisor bounds)**.** *Let $\tau(n)$ be the number of positive divisors of the positive integer n. For every $\eta > 0$, $\tau(n) \ll_\eta n^\eta$. For $V > 0$, the number of ordered integer pairs satisfying $x^2 + y^2 = V$ is at most $4\tau(V)$.*

Proof. For all sufficiently large primes p and every integer $a \ge 1$, $a + 1 \le 2^a \le p^{\eta a}$. For each of the finitely many smaller primes, $\sup_{a \ge 0} (a+1)p^{-\eta a} < \infty$. Multiplying over the prime factorization proves the divisor bound.

The Gaussian integers are Euclidean for the norm: after dividing in $\mathbb{C}$, round both coordinates to obtain a remainder of norm at most half the divisor's norm. The Euclidean algorithm gives a greatest common divisor and Bézout's identity. An irreducible is therefore prime, and induction on the norm gives unique factorization. If a rational prime p is reducible in $\mathbb{Z}[i]$, its factors have norm p, since their integer norms multiply to p^2; thus p is associated to $\pi\bar{\pi}$. The two factors may be associates. If p remains irreducible, its exponent in an element of norm V is determined by the exponent of p in V, when such an element exists. The same is true in the associate case. If $\pi, \bar{\pi}$ are nonassociate and $p^a \| V$, there are at most $a + 1$ ways to distribute their exponents. The four units give at most $4 \prod_{p^a \| V} (a + 1) = 4\tau(V)$ elements $x + iy$ of norm V. □

Lemma 4.2 (Fixed quadratic)**.** *Fix $Q(t) = at^2 + bt + c \in \mathbb{Z}[t]$ with $a \ne 0$. For $N \ge 2$ and any $B \subseteq [1, N] \cap \mathbb{Z}$, the number of ordered solutions $(u, v) \in B^2$ of $Q(u) + Q(v) = s$ is $O_{Q,\eta}(N^\eta)$, uniformly in the integer s. If instead B is any set of integer parameters whose values lie in $[0, X]$, $X \ge 2$, that number is $O_{Q,\eta}(X^\eta)$, independently of a parameter cutoff. Every value of Q has at most two parameter preimages.*

Proof. The identity

$$4aQ(t) = (2at + b)^2 + 4ac - b^2 \tag{4.1}$$

maps a solution injectively to $x = 2au + b$, $y = 2av + b$ satisfying

$$x^2 + y^2 = 4as + 2b^2 - 8ac =: V. \tag{4.2}$$

For $V < 0$ there is no solution; for $V = 0$ there is at most one; for $V > 0$ Lemma 4.1 bounds their number by $4\tau(V)$. When an original solution exists, $|s| \ll_Q N^2$ and hence $|V| \ll_Q N^2$. Apply the divisor bound with exponent $\eta/2$. In the physical-size setting $0 \le s \le 2X$, so $|V| \ll_Q X$ and the same proof applies with a suitably small divisor exponent. Finally, $Q(t) - z$ has at most two roots for any z. □

4.2 Fixed cubics and their zero branches

Lemma 4.3 (Fixed cubic)**.** *Fix $Q(t) = at^3 + bt^2 + ct + d_0 \in \mathbb{Z}[t]$ with $a \neq 0$. For $N \ge 2$ and $B \subseteq [1, N] \cap \mathbb{Z}$, the number of ordered solutions $(u, v) \in B^2$ of $Q(u) + Q(v) = s$ is $O_{Q,\eta}(N^\eta)$, uniformly in the integer s. If B is any set of* positive *integer parameters with $Q(B) \subseteq [0, X]$, $X \ge 2$, the number is $O_{Q,\eta}(X^\eta)$, independently of a parameter cutoff. Every value has at most three parameter preimages.*

Proof. Put

$$A = 9ac - 3b^2, \qquad B_0 = 2b^3 - 9abc + 27a^2d_0. \tag{4.3}$$

Direct expansion gives

$$27a^2Q(t) = (3at + b)^3 + A(3at + b) + B_0. \tag{4.4}$$

For $x = 3au + b$, $y = 3av + b$, the pair-sum equation implies

$$V := 27a^2s - 2B_0 = (x + y)(x^2 - xy + y^2 + A). \tag{4.5}$$

The affine maps to x and y are injective.

If $V \neq 0$, let $\ell = x + y$. This is a signed divisor of V; there are $2\tau(|V|)$ possibilities. For fixed ℓ,

$$xy = \frac{\ell^2 + A - V/\ell}{3}. \tag{4.6}$$

The sum and product determine at most two ordered root pairs. Thus there are at most $4\tau(|V|)$ solutions. For parameters at most N and an existing solution, $|V| \ll_Q N^3$. For selected values in $[0, X]$, one has $s \in [0, 2X]$ and $|V| \ll_Q X$. Lemma 4.1, with the exponent decreased as needed, gives both asserted estimates when $V \neq 0$.

If $V = 0$, the first branch in (4.5) is $x + y = 0$. It forces

$$u + v = -\frac{2b}{3a}. \tag{4.7}$$

If the right side is a positive integer H, there are at most $H - 1$ ordered positive pairs; otherwise there are none. This is a constant depending only on the fixed polynomial. The other branch is $x^2 - xy + y^2 = -A$. Since

$$2(x^2 - xy + y^2) = (x - y)^2 + x^2 + y^2, \tag{4.8}$$

it lies in the fixed disk $x^2 + y^2 \le -2A$ and has $O_Q(1)$ integer points. For $A > 0$ it is empty. These branches cover $V = 0$; their intersection causes only overcounting. The restriction to positive parameters and the fixed coefficients are both used. The three-preimage bound follows from the degree. □

For a set of distinct selected values, choose one parameter preimage per value. This injects ordered value pairs into parameter pairs. Hence Lemmas 4.2 and 4.3 bound (3.1) for any subset of the selected values.

4.3 Coefficient-uniform quadratics

Lemma 4.4 (Three consecutive values)**.** *Under (1.6), the pair-sum multiplicity of $Q(I)$ is $O_\eta(X^\eta)$, with a constant independent of the coefficients and of I. The same holds for every subset of $Q(I)$, and every value has at most two parameter preimages.*

Proof. If $|I| \le 2$, there are at most four ordered parameter pairs. Otherwise let m_0 be the left endpoint and write $q(j) = Q(m_0 + j) = Aj^2 + Bj + C$. The integers $j = 0, 1, 2$ belong to the translated interval. Set $y_j = q(j)$ for these three integers. Then

$$A = (y_2 - 2y_1 + y_0)/2, \quad B = y_1 - y_0 - A, \quad C = y_0, \tag{4.9}$$

$$|A| \le X/2, \qquad |B| \le X, \qquad |C| \le X. \tag{4.10}$$

The translated polynomial has integer coefficients and $A \neq 0$. If $q(j) + q(k) = s$ with parameters in this interval, then $s \in [X, 2X]$. Its quadratic norm parameter satisfies

$$|4As + 2B^2 - 8AC| \le 10X^2. \tag{4.11}$$

Use the cases $V < 0$, $V = 0$, $V > 0$ from (4.2). In the last case the divisor bound with exponent $\eta/2$ gives a constant depending only on η. Passing to distinct value pairs, and then to subsets, decreases the count. Each value has at most two preimages by the degree. □

The interval condition supplies the three consecutive values. The proof allows an interval crossing the vertex and a negative leading coefficient. For example $q(j) = -j^2 + 2j + 1$ on $j = 0, 1, 2$ has values $1, 2, 1$ in $[1, 2]$, including a repeated value.

Proposition 4.5 (Moving cubic multiplicities)**.** *For integers $M \ge 3$ let*

$$Q_M(t) = 2(2t - M - 1)^3 + 6M^3, \qquad X = 8M^3. \tag{4.12}$$

The values $Q_M(1), \dots, Q_M(M)$ are even, positive, distinct, and belong to $[X/2, X]$. Their pair-sum multiplicity at $12M^3$ is at least M. Consequently a coefficient-independent bound $O_\eta(X^\eta)$ for these multiplicities fails for every $0 < \eta < 1/3$.

Proof. For $1 \le t \le M$, $z_t = 2t - M - 1$ lies in $[-(M-1), M-1]$. Thus

$$4M^3 \le 6M^3 - 2(M-1)^3 \le Q_M(t) \le 6M^3 + 2(M-1)^3 \le 8M^3.$$

The cubic is strictly increasing, and the values are even. Since $z_{M+1-t} = -z_t$, their sum is $12M^3$ for all M ordered choices of t. Finally $M/(8M^3)^\eta \to \infty$ when $\eta < 1/3$. □

This construction exhibits the growing linear zero branch for moving cubic coefficients. Its conclusion concerns the uniform multiplicity extension of Lemma 4.4. The constant for a fixed cubic in Lemma 4.3 is allowed to depend on Q.

5 The Li–Liu consequences

We make explicit the Fourier coefficients to which the transfer is applied. By finite orthogonality,

$$\int_{\mathbb{T}} S_X(\alpha)^2 e(-n\alpha)\,d\alpha = \sum_{p_1+p_2=n} (\log p_1)(\log p_2) w(p_1/X) w(p_2/X), \tag{5.1}$$

$$\int_{\mathbb{T}} |S_X(\alpha)|^2 e(-h\alpha)\,d\alpha = \sum_{p_2-p_1=h} (\log p_1)(\log p_2) w(p_1/X) w(p_2/X). \tag{5.2}$$

The second identity assigns the positive exponent to p_2. In both sums the support of w places the primes below X. For a value having no prime-sum representation, (5.1) is zero. For a value having no prime-difference representation with both primes at most X, (5.2) is zero. These implications supply the vanishing conditions in Lemma 3.1.

5.1 Physical blocks for sums

Assume the inputs of Theorem 1.1. Fix Y above the analytic threshold and let $\mathcal{A}$ consist of the distinct even values $Q(m) \in (Y/2, Y]$, $m \geq 1$, that are not sums of two primes. The physical forms of Lemmas 4.2 and 4.3 give

$$r_{\mathcal{A}}(v) \ll_{Q,\eta} Y^{\eta}. \tag{5.3}$$

Use S_Y with the common κ of Inputs 2.3 and 2.4. Equations (2.9), (3.6), and (2.4) verify all the conditions of Corollary 3.2. Hence

$$|\mathcal{A}| \ll_{Q,\Theta,\varepsilon} Y^{1/4+\varepsilon}. \tag{5.4}$$

There are at most d parameters per value, so the same bound holds for parameters on this physical block.

Proof of the sum assertion in Theorem 1.1. For $1 \leq m \leq N$, $|Q(m)| \leq A_Q N^d =: T$. Use the disjoint physical blocks

$$(T2^{-j-1}, T2^{-j}], \qquad j \geq 0. \tag{5.5}$$

For fixed $\gamma > 0$, apply (5.4) with exponent $1/4+\gamma$ to the blocks above the analytic threshold. Their sum is

$$\ll_{Q,\Theta,\gamma} \sum_{j\geq 0} (T2^{-j})^{1/4+\gamma} \ll_{Q,\Theta,\gamma} T^{1/4+\gamma}.$$

Values below the threshold form a fixed finite set of integers, each with at most d preimages, so they contribute $O_{Q,\Theta}(1)$. Choose $\gamma > 0$ so that $d\gamma \leq \varepsilon$ and substitute $T = A_Q N^d$. This gives the sum bound in (1.4). □

5.2 One global scale for differences

Proof of the difference assertion in Theorem 1.1. Take the prime scale $Z = C_Q N^d$. Every positive even value $h = Q(m)$ with $m \leq N$ lies in $[2, Z/2]$, by (1.3). Let $\mathcal{A}$ consist of those distinct values which fail to have a prime-difference representation with both primes at most Z. Use S_Z, $F = |S_Z|^2$, and the same Li–Liu arcs. Equation (5.2) gives the vanishing full coefficient, and (2.10) gives its major lower bound. The physical multiplicity lemmas apply to selected values in $[0, Z]$ and give $r_{\mathcal{A}}(v) \ll_{Q,\eta} Z^{\eta}$ for the entire set. Corollary 3.2 yields $|\mathcal{A}| \ll_{Q,\Theta,\gamma} Z^{1/4+\gamma}$. Multiply by at most d preimages per value and choose $d\gamma \leq \varepsilon$. The bounded range of Z below the analytic threshold is absorbed by increasing the constant. □

The smooth sum delivers primes strictly below Z. The common statement uses $\le Z$ to include the endpoint convention in the alternative route. An integer having no unrestricted prime-difference representation belongs to this bounded exceptional set as well.

5.3 Uniform quadratic families

Proof of (1.7). Assume first that X exceeds the analytic threshold. For sums take the distinct exceptional even values in $Q(I)$, use S_X, and replace (5.3) by Lemma 4.4. Corollary 3.2 gives $O_{\Theta,\varepsilon}(X^{1/4+\varepsilon})$ values, uniformly in the coefficients. Each value has at most two parameter preimages. For differences take prime scale $2X$. All target values lie in $[(2X)/4, (2X)/2]$, within the range of Input 2.4. The same multiplicity estimate and the difference transfer give the same count, and the Fourier support supplies primes below $2X$.

For bounded X there are at most $\lfloor X \rfloor + 1$ integer values in $[X/2, X]$, each with at most two preimages. Their total is absorbed by a constant depending only on the fixed analytic threshold and ε, uniformly in Q and I. This includes empty intervals and those of length at most two. □

6 The Pintz–MPR alternative

The minor-arc estimate must hold for the same lower-truncated prime sum T_X and the same P as the major coefficients in Input 2.5.

Lemma 6.1 (Both prefixes on the same minor arcs)**.** *Fix the parameters of Input 2.5. For its chosen P,*

$$\sup_{\alpha \in \mathfrak{m}_X(P)} |T_X(\alpha)| \ll_\eta X^{19/24+\eta} \tag{6.1}$$

for every $\eta > 0$, assuming Input 2.6.

Proof. We have

$$X^{4/9-2\nu} \le P \le X^{4/9-\nu}, \qquad 4/9 - 2\nu > 5/12. \tag{6.2}$$

Fix $\alpha \notin \mathfrak{M}_X(P)$. For each $Y \in \{X, X_1\}$ choose a Dirichlet approximation a/q with $q \le \sqrt{Y}$ and $|\alpha - a/q| \le 1/(q\sqrt{Y})$, reducing the fraction if needed. The approximation is taken modulo one in (2.3). If $q > P$, then $B_Y \ge q > P \ge PY/X$. If $q \le P$, exclusion from the same major arcs gives $|\alpha - a/q| > P/(qX)$, and again $B_Y > PY/X$. Thus

$$\frac{Y}{\sqrt{B_Y}} < \sqrt{\frac{XY}{P}} \le \frac{X}{\sqrt{P}} \le X^{7/9+\nu} \le X^{19/24}. \tag{6.3}$$

Apply (2.16) to both prefixes, changing the strict endpoint if needed, and subtract. Since $Y \le X$, both the $Y^{19/24}$ term and the arbitrary small power are bounded at scale X. The total weight of proper prime powers is

$$\sum_{\substack{p^j \le X \\ j \ge 2}} \log p \ll X^{1/2} (\log X)^2. \tag{6.4}$$

Indeed there are $O(\log X)$ possible exponents, at most $\sqrt{X}$ bases for each, and weight at most $\log X$ per term. Removing these terms leaves T_X and costs less than the bound in (6.1). □

The small constant ν is chosen admissibly for the major-arc formula before X varies. As a rational illustration of the exponent comparison, $\nu = 1/144$ gives $P_0 = X^{7/16}$, $P \ge X^{31/72}$, and $X/\sqrt{P} \le X^{113/144} < X^{19/24}$. The proof only requires the strict inequality in (6.2).

Lemma 6.2 (Alternative physical block)**.** *Under the inputs of Theorem 1.2, for a fixed quadratic or cubic Q, the number of positive integer parameters with even $Q(m) \in (Y/2, Y]$ unrepresented as a prime sum is $O_{Q,\Theta,\varepsilon}(Y^{7/24+\varepsilon})$. The number lacking a prime-difference representation with both primes at most $2Y$ has the same bound. For a quadratic family satisfying (1.6) at scale Y, both bounds have constants independent of the coefficients and the parameter interval.*

Proof. For sums use T_Y, its arcs from Input 2.5, and the distinct exceptional values. The full Fourier coefficient is zero, the major real part is at least cY, and (3.6) gives $B_2 \le Y(\log Y)^2$. Lemma 6.1 gives $M \ll_\eta Y^{19/24+\eta}$. The physical polynomial multiplicity lemmas and the second part of Corollary 3.2 give the desired bound for values; the degree bounds their parameter preimages.

For differences use prime scale $X = 2Y$. Every target $h \in (Y/2, Y]$ lies in $[X/4, X/2]$, exactly the range used in (2.15). Failure of a representation with both primes at most X makes the full coefficient of $|T_X|^2$ vanish. The same transfer gives the asserted count and prime bound. For uniform quadratics use Lemma 4.4 instead of the fixed-coefficient lemmas. Bounded scales are covered by the number of integer values and the degree bound, as in the proof of (1.7). □

Proof of Theorem 1.2 and (1.8). For fixed Q sum Lemma 6.2 over (5.5) with $T = A_Q N^d$. The geometric sum is $O_{Q,\Theta,\gamma}(T^{7/24+\gamma})$ for either sign. For each difference block its delivered primes have size at most $2T2^{-j} \le 2A_Q N^d < C_Q N^d$. Every parameter failing the global prime bound therefore belongs to a block failure just counted. Values below the fixed analytic threshold give boundedly many parameters. Choose $d\gamma \le \varepsilon$ to obtain (1.5). For a uniform quadratic family apply the physical lemma at $Y = X$, giving primes at most $2X$ and (1.8). □

Proof of Corollary 1.4. Input 2.2 supplies Hypothesis 2.1 at $\Theta = 7/8$, below both required boundaries. Apply the preceding theorems with this zero-free input and the remaining inputs specified in each theorem. □

A Ancillary checks

The ancillary programs check the polynomial identities (4.1), (4.4), and (4.8); the rational parameter choice recorded in (2.12); and the exponent comparisons in the finite transfer and in (6.3). The program `anc/exact_checks.py` checks the identities and the fixed and moving cubic examples. The program `anc/interface_checks.py` checks the coefficient reconstruction, the power comparisons, and the change from 19/24 to 7/12 when the Perelli–Pintz sum is restricted to the exceptions. The program `anc/verify_claims.py` additionally enumerates pair-sum multiplicities of fixed quadratics and cubics on finite parameter ranges and checks the three-value bounds on integer samples.

These are exact finite checks of the algebra and examples used here. The deductions for general parameters are proved in Sections 3–6; the analytic theorems are the inputs of Section 2. The ancillary README gives the commands, dependencies, and scope of each check.

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Use of AI

The proofs and the text of this paper were produced with AI systems (GPT-6.1 Sol, GPT-6 Astra and Claude Opus 5.5) under the author's direction, checked by independent reviews with AI models and by the computations in the ancillary files. No human expert has reviewed them. The author takes responsibility for the content.